

# **From Laboratory Model to Statewide Deployment: Continuous Enhancement of a Wrong-Way Driving Detection System from 10 to 200 Existing CCTV Cameras**

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## ABSTRACT

**Objectives:** Vision-based wrong-way driving (WWD) detection has been investigated on controlled evaluation sets, yet no state agency operates such a system at network scale. This paper reports the continuous enhancement of one published laboratory model into a system operating across 294 Iowa DOT CCTV cameras. The objective throughout was not algorithmic sophistication but the highest detection performance per dollar that a department of transportation can implement today using cameras it already owns. The objective is to provide a model for a traffic operator decision-support system that minimizes missed-event risk, false-alert burden, verification delay, operator workload, and computational cost across a statewide camera network.

**Methods:** We compare four deployment architectures on cost (per-camera edge compute, dedicated-sensor retrofit, cloud, and the hybrid on-premises design we adopted) and document a continuous series of twelve system enhancements developed over [N months] of operation. The three most recent enhancements, currently in progress, automate roadway filter generation, route candidate alerts according to the expected reduction in operator decision risk per unit of delay and computation, and apply multimodal review only when additional visual evidence is valuable. Evaluation uses production alerts, confirmed WWD incident records, and held-back historical clips, with emphasis on false-alert reduction, event retention, operator burden, latency, and cost.

**Findings:** The hybrid architecture costs an estimated \$125 per camera-year over five years, compared with approximately \$750 for edge compute, \$4,500 to \$9,800 for dedicated-sensor retrofit, and \$120 to \$320 for cloud processing. **[After reconciling the evaluation files, report: production-alert precision, confirmed-event retention, false-alert reduction, median operator review time, detection-to-dispatch latency, and compute/API cost. Do not insert the provisional 68%, 89%, or 72% values until their numerators and denominators are verified.]**

**Novelty:** This is the first published account of CCTV-based WWD detection operated continuously at a scale exceeding 100 cameras, the first deployment-economics comparison of edge, retrofit, cloud, and hybrid architectures for this task, and the first evaluation of a safety-asymmetric, cost-aware evidence-routing policy that reserves multimodal review for uncertain alerts and returns structured evidence to traffic operators.

**Practical Applications:** A department of transportation can replicate this system on cameras it already owns using commodity hardware and published components. The cost model and the LLM enhancements reduce per-camera setup and review labor by [N%], making deployments of 200 or more cameras operationally sustainable.

## INTRODUCTION

Wrong-way driving (WWD) crashes are rare but disproportionately severe. Between 20% and 25% of WWD crashes are fatal, compared with roughly 0.5% for vehicle crashes overall (1). Between 2015 and 2018, the United States recorded approximately 500 fatalities per year from WWD crashes on divided highways, a 34% increase over the preceding period (2). Iowa has experienced the same trend; a study of a 25-mile section of U.S. Route 30 between Nevada and Boone documented 87 WWD incidents from 2014 to 2017 (3). Nearly 60% of WWD events involve drivers with a blood alcohol concentration of 0.08% or higher (1, 4), which means the driver is unlikely to self-correct. Detection and rapid response are therefore the operative countermeasures.

Conventional countermeasures, including static signage, pavement markings, ramp geometry modifications, and spot detection systems using inductive loops, radar, or thermal cameras, require capital investment at every covered site and protect only the instrumented location. State departments of transportation, however, already operate large CCTV networks installed for surveillance; Converting this existing infrastructure into a WWD detection network requires no new roadside hardware, but it does require detection software that can operate unattended, at scale, for extended durations

This paper is structured as a study of the transition from laboratory model to statewide deployment. Its starting point is a published laboratory result: a self-calibrating deep learning WWD detector that achieved 0.99 precision and 0.97 recall on 10 cameras (5). That study resolved the accuracy question but, like most published WWD detection studies, did not address the operational question. When we extended the system toward statewide deployment, the failures that consumed the following two years were not model failures. They included RTSP streams that disconnected without warning, pan-tilt-zoom (PTZ) cameras that shifted between similarity checks, calibration ranges learned from a single low-traffic day, intersections where legal turning movements resembled violations, operator alert fatigue, and processor saturation at 110 streams. These failure modes do not appear in the WWD detection literature, and their absence helps explain why no state currently operates such a system at network scale.

The system described here was therefore developed through a continuous series of enhancements, beginning with the laboratory model and proceeding through stream reliability, calibration logic, camera-shift monitoring, false-positive management, and computational scaling, up to the three enhancements currently in progress, which employ a multimodal large language model (LLM) to remove the remaining per-camera human labor. We are deliberate about the design philosophy: the detector is a pre-trained ResNet and the tracker is Kanade-Lucas-Tomasi (KLT), selected because they process  $[N]$  streams per GPU rather than because they lead any benchmark. Every design decision optimizes detection performance per dollar under the constraint that a department of transportation can implement the system today, on cameras it already owns, with hardware available off the shelf. We make this trade explicit by costing four deployment architectures and by measuring the per-camera human labor that the LLM enhancements are designed to remove.

The contributions of this paper are as follows:

- **Operational record.** The first published account of CCTV-based WWD detection operated continuously at a scale exceeding 100 cameras, including the previously unpublished series of enhancements that carried the laboratory model from 10 to **[200, CONFIRM]** cameras (Section 5).
- **Deployment economics.** A cost comparison of edge, retrofit, cloud, and hybrid architectures for network-scale WWD detection, with all assumptions stated, showing why the

hybrid design provides the highest detection performance per dollar for a mid-size agency (Section 4).

- **Decision-support enhancements.** Three enhancements in progress that extend the series: automated region-of-interest (ROI) filter generation validated by IoU against manual annotation, risk- and cost-aware evidence routing, and multimodal verification with an operator-ready evidence package (Section 5.7).
- **Operational evaluation.** Adjudication of 319 production candidate clips, reconciliation against 275 confirmed WWD incident records, and comparison of production, always-review, and routed-review policies on event retention, false-alert burden, delay, operator workload, and cost (Sections 6 and 7).

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the production system. Section 4 presents the deployment architecture comparison. Section 5 documents the enhancement series, including the three operator-centered enhancements in progress. Section 6 defines the evaluation framework, Section 7 reports results, Section 8 discusses limitations, and Section 9 concludes.

## RELATED WORK

### Dedicated-Sensor Detection and State DOT Deployments

Arizona DOT established the feasibility of dedicated-sensor WWD detection a decade ago through proof-of-concept testing of five off-the-shelf technologies, all of which produced missed and false calls (6); this work led to the corridor-scale concept of operations behind the I-17 pilot (7). The most comprehensive evaluation to date, conducted by the Texas A&M Transportation Institute for TxDOT across ten commercial systems, reached a conclusion central to the present study: all systems detected wrong-way vehicles, no technology universally outperformed the others, and false calls were the dominant problem, with one optical system logging approximately 95% false calls caused by shadows and headlight reflections (8). The most rigorous hardware evaluation in the field thus identifies false positives, rather than detection capability, as the binding constraint.

Deployment experience confirms both the effectiveness and the cost structure of the dedicated-sensor approach. Tennessee DOT pilot installations cost \$15,769 to \$45,000 per off-ramp, with maintenance up to \$3,600 per site-year (9). The Arizona I-17 corridor, comprising 90 thermal cameras over 15 miles, cost \$3.7 million to install; detections preceded 911 calls by an average of 98 seconds, and 88% of drivers self-corrected (10, 11). Caltrans monitored 12 ramps for 39 months using solar-powered vision units (12), and Florida reports more than 80% self-correction upon sign activation in its statewide program (13). Detection-activated countermeasures measurably outperform static signage (14) and produce positive benefit-cost outcomes on instrumented networks (15, 16). The limitation of this approach is structural rather than technical: capital scales linearly with the number of covered locations, which is why agencies instrument only the highest-risk ramps.

### Vision-Based Detection on Existing CCTV

Camera-based research has progressed from optical-flow and background-subtraction methods that learn a dominant motion direction for each image region (17, 18, 19) to deep learning pipelines, including rule-based violation detection over detector output (20), YOLO-family detection with centroid or Kalman tracking (21, 22), appearance-based classification of vehicle front and rear views that infers direction without tracking (23), and automatic lane-boundary extraction feeding

YOLO-based direction validation (24). Recent studies continue this pattern with newer detectors and trackers (25, 26). Across this literature, evaluation is conducted on single cameras or small sets of curated clips, and no study reports operation on a camera network. The largest published operational WWD detection dataset, comprising 726 detections across 48 toll ramps over multiple years, was collected with dedicated radar, laser, and thermal sensors rather than cameras (27). No vision-based WWD detection algorithm paper has appeared in Transportation Research Record in the past five years.

Our own prior work provides the laboratory result from which this paper proceeds: a prototype on Iowa DOT CCTV (28) refined into an end-to-end self-calibrating model that treats PTZ orientation as a variable and recalibrates after camera rotation, achieving 0.99 precision and 0.97 recall (5). As with the remainder of the vision-based literature, that evaluation was short in duration and small in scale. To our knowledge, no published study reports months of continuous WWD detection operation across a network of more than 100 cameras, the deployment-architecture economics of such operation, or the false-positive burden carried by operations staff. **[Claim verified against literature through July 2026; re-run the check the week of submission]**

### **LLMs and Vision-Language Models in Traffic Video**

Multimodal LLMs are being applied to traffic video along several directions: surveys of vision-language model (VLM) approaches to traffic anomaly detection (29), VLM anomaly scoring with verbalized explanations (30), zero-shot and few-shot safety-event detection, in which accuracy near 79% indicates that an unassisted multimodal model is not deployable as a primary detector (31), and LLM-summarized city-scale camera analytics, most prominently across approximately 1,000 New York City cameras, although that system processes low-frequency snapshots for congestion monitoring rather than per-alert safety detection (32). Cascades that pass detector output to a VLM and then to an LLM have been proposed for incident reporting, with the language model enriching rather than filtering alerts (33). The cost and latency of VLM video processing at multi-camera scale is an active research concern (34).

Two directions border the present contributions. First, commercial surveillance software now offers VLM-based alert verification to reduce false positives in general settings such as personal protective equipment compliance (35). The cascade pattern therefore exists as a product feature; however, we find no peer-reviewed evaluation of VLM alert review in a live, safety-critical traffic detection deployment, and no published field results on operator-facing false-alert rates. Second, promptable open-vocabulary segmentation (36) can produce roadway masks from text prompts, but no published work uses a multimodal LLM to author detection-zone geometry consumed by a production traffic pipeline and validated against human annotation. A recent review of 65 studies of LLMs in transportation identifies deployment and scalability as the open gaps in the field (37), and a survey of LLM-based crash detection in video reaches the same conclusion (38). We therefore evaluate three claims separately: automated roadway-filter generation reduces setup labor; risk- and cost-aware routing reduces expensive review without sacrificing event retention; and structured multimodal verification reduces operator false-alert burden while preserving uncertain safety events for human review.

### **The Gap Between Laboratory Validation and Deployment**

The gap documented in this paper is well characterized outside transportation. The model constitutes a small fraction of a production machine learning system (39), and deployment failures

cluster in data infrastructure, integration, monitoring, and drift rather than in the model itself (40). Transportation has its own precedent: a nationwide survey found that most traffic management centers had abandoned, or had never operationalized, automated incident detection algorithms, with excessive false alarm rates a leading cause (41, 42). Research on clinical monitoring explains the mechanism, in that responders desensitize when 80% to 99% of alarms are false (43). The systems literature on video analytics identifies per-camera compute and network cost as the binding constraint once deployments reach hundreds of cameras (44). Prior work by this group on the same Iowa network, including CNN-based congestion detection on live CCTV (45), cloud video-analytics cost feasibility analysis (46), and real-time crash detection from connected-vehicle data (47), follows the same trajectory. The present paper provides the WWD instance of this documented gap, with the operational record made public.

### **SYSTEM OVERVIEW: FROM LABORATORY MODEL TO PRODUCTION PIPELINE**

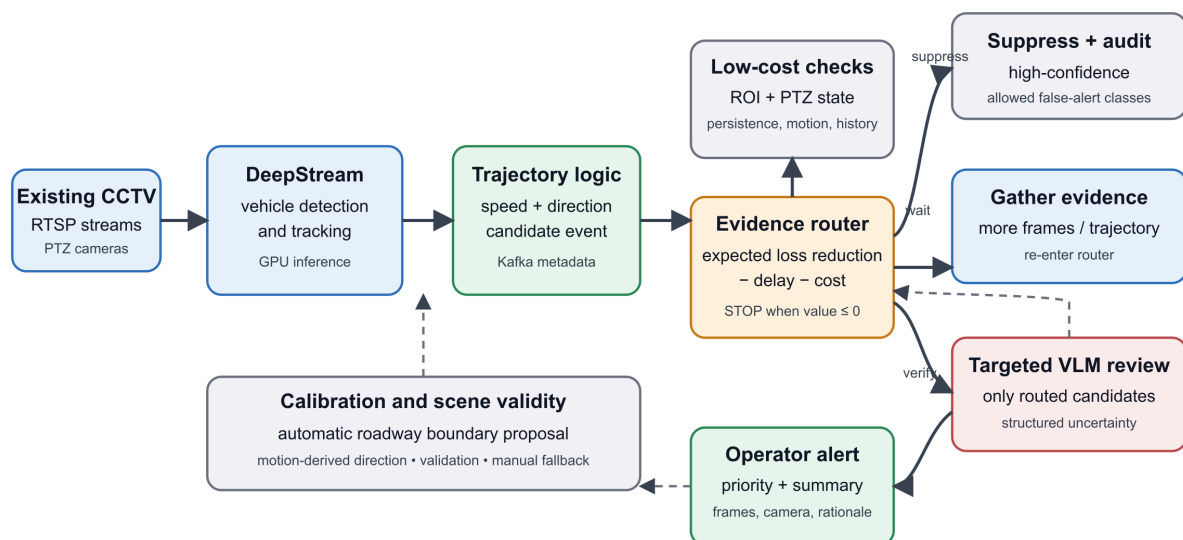
The production system retains the central idea of the laboratory model, namely unsupervised, self-calibrating direction learning that requires no manual per-camera setup, and surrounds it with the operational components that the laboratory version did not require. The system ingests RTSP live streams from Iowa DOT CCTV cameras into an NVIDIA DeepStream pipeline. A pre-trained ResNet model detects vehicles, and the KLT tracker maintains vehicle identities across frames, yielding position, speed, and direction for each vehicle at each timestamp. Metadata flows through Apache Kafka, with clustered consumers and partitioned topics, to a post-processing tier that applies the detection logic. Processing is divided between a GPU machine, which performs video inference and Kafka production, and a CPU machine, which performs post-processing, anomaly detection, and alert generation. This division was adopted when scaling exposed CPU saturation, as described in Section 5.5.

Calibration proceeds independently for each camera. One day of vehicle trajectories is collected, and DBSCAN clusters the trajectories, weighted by x-intercept, angle, and median point, to define the two directional regions of interest. These regions are extended to the road edges by k-nearest-neighbor (KNN) assignment and closed with a convex hull. The camera field of view is discretized into a 50 by 50 grid, and the permitted speed and direction for each grid cell are computed from observed traffic, with KNN interpolation filling cells that received no observations. A vehicle is flagged as traveling the wrong way when its direction deviates from the permitted range, defined as the mean plus or minus 45 degrees, for more than 90 frames. A vehicle is flagged as stalled when its speed remains near zero for 1,000 or more consecutive frames. Upon a confirmed detection, the system assembles a 60-second clip from a rolling 1,500-frame buffer and transmits it to operators by email.

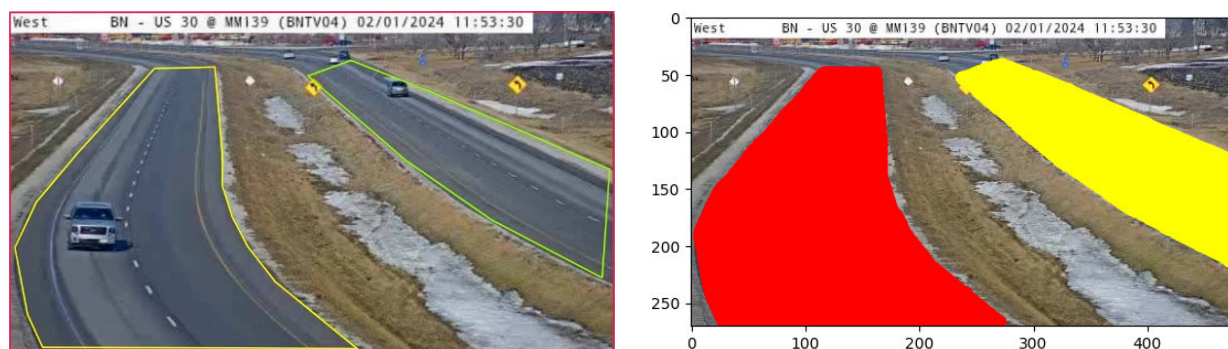
Two guard subsystems address the dominant false-positive sources. Camera orientation is verified by optical character recognition (OpenCV EAST with Tesseract) applied to the direction text overlaid in the video header. Camera shift, zoom, and focus changes are detected by LPIPS image similarity against a calibration reference frame with a threshold of 0.5, and WWD detection is paused until the view stabilizes. Road segments generating more than 10 wrong-way flags per hour are automatically pixel-blocked.

### **DEPLOYMENT ARCHITECTURE AND COST CONSIDERATIONS**

The governing design question throughout this project was not which architecture maximizes accuracy, but which architecture provides the most detection coverage per dollar using cameras



**FIGURE 1** End-to-end architecture of the deployed WWD pipeline and the proposed operator-centered evidence-routing layer. Continuous DeepStream and trajectory processing remains on premises; the router invokes additional evidence or multimodal review only when its expected decision value exceeds its delay and cost.



**FIGURE 2** Self-calibration artifacts from the deployed pipeline: clustered directional roadway regions (left) and the interpolated permitted speed-and-direction field that is used to evaluate vehicle trajectories (right).

the agency already owns. Four candidate architectures were costed before the present design was adopted. The comparison below uses [**state amortization period, labor rate, cloud pricing basis, camera count N**]; all assumptions are stated so that another agency can substitute its own values.

### The Four Candidate Architectures

- **Per-camera edge compute.** A GPU-capable edge device is installed at each camera or camera hub. This architecture minimizes backhaul bandwidth and latency, but capital scales linearly with the number of cameras. A Jetson Orin NX 16GB module alone costs \$999 at 2026 pricing (48), and a deployed unit, including carrier board, hardened enclosure, power, cabinet space, and field installation, reaches approximately \$3,000 per site, or approximately \$600,000 for 200 cameras, in addition to the maintenance burden of 200 roadside devices. Because PTZ surveillance cameras rotate on operator demand, edge devices also inherit every recalibration problem locally.
- **Dedicated-sensor retrofit.** Commercial WWD systems, typically radar, LiDAR, or thermal units mounted at ramps with active warning signs, cost \$18,000 to \$45,000 per off-ramp installed in Tennessee DOT pilot deployments (9), with annual maintenance up to \$3,600 per site. The Arizona I-17 thermal corridor cost \$3.7 million for 15 miles and 90 thermal cameras, approximately \$41,000 per camera (10, 11). These systems provide excellent point accuracy, but coverage equals the set of instrumented sites, so protecting 200 locations requires \$3.6 to \$9.0 million in capital alone. The present approach is an alternative to this architecture rather than a variant of it.
- **Cloud processing.** All RTSP feeds are streamed to cloud GPU instances. This architecture requires no capital outlay and scales elastically. Approximately 200 streams require on the order of six T4-class instances at \$0.23 to \$0.53 per hour each (49), or approximately \$12,000 to \$28,000 per year, plus a dedicated upstream circuit of at least 400 Mbps from the DOT network, or approximately \$12,000 to \$36,000 per year. The raw cost is competitive with the hybrid design; the binding constraints are instead the perpetual recurring cost, backhaul latency, and DOT video-security and retention policies that complicate off-premises processing [**cite Iowa DOT policy constraint if applicable**].
- **Hybrid on-premises (adopted).** Central commodity GPU servers perform inference on all streams, and a separate CPU server performs post-processing. This division was adopted when scaling exposed CPU saturation (Section 5.5). Capital consists of a small number of servers, approximately \$75,000 for a dual-GPU inference server and a CPU post-processing server [**REPLACE with actual BOM**], rather than 200 field devices. The marginal cost of adding a camera is near zero until a GPU saturates, and routed multimodal review shifts only the uncertain-alert workload to a metered API. The predecessor project measured 160 streams on two NVIDIA T4 GPUs at less than one second of perception latency, approximately 65% GPU memory use, and 69–71% GPU utilization (28). Its five-day Google Cloud deployment cost \$21.59 per day for 160 cameras with sustained-use pricing, or \$49.30 per camera-year in 2020 dollars; GPU, vCPU, and remaining resources accounted for 46.4%, 19.7%, and 33.9% of cost, respectively (28). These historical measurements are reported as an empirical scaling baseline, not as current 2026 prices.

Two observations from Table 1 merit emphasis. First, retrofit and hybrid architectures are not direct substitutes: retrofit systems actuate warning signs at the ramp, whereas camera-based detection alerts the traffic management center. An agency pursuing both functions would deploy



**TABLE 1 Deployment-Architecture Cost Comparison for 200-Camera WWD Detection**

| Cost component            | Edge                        | Retrofit                        | Cloud                       | Hybrid (ours)                                  |
|---------------------------|-----------------------------|---------------------------------|-----------------------------|------------------------------------------------|
| Capital (200 cameras)     | ≈ \$600K (est.)             | \$3.6M–\$9.0M (9)               | ≈ 0                         | ≈ \$75K [BOM]                                  |
| Recurring / yr            | ≈ \$30K field maint. (est.) | up to \$720K (≤\$3.6K/site (9)) | \$24K–\$64K (GPU + circuit) | ≈ \$10K power/rack + LLM API [TO FILL]         |
| Marginal cost, camera N+1 | ≈ \$3K (new device)         | \$18K–\$45K (full site)         | ≈ \$60–\$140/yr (GPU share) | ≈ 0 until GPU saturates (~100 streams)         |
| Setup labor / camera      | field install + config      | full construction               | config only                 | [TO FILL: hrs before vs. after Enhancement 10] |
| 5-yr total, 200 cameras   | ≈ \$750K                    | ≈ \$4.5M–\$9.8M                 | ≈ \$120K–\$320K             | ≈ \$125K + LLM API                             |
| 5-yr \$ per camera-yr     | ≈ \$750                     | ≈ \$4,500–\$9,800               | ≈ \$120–\$320               | ≈ \$125                                        |

*Note:* 2026 USD estimates assume a 5-yr horizon, ~2 Mbps per stream, and ~100 streams per modern inference GPU. T4-class cloud instances use reserved-to-on-demand pricing (49); edge uses a \$999 Orin NX module plus ~\$2K carrier, enclosure, and installation per site (48); retrofit uses the TDOT per-ramp range (9), corroborated by ADOT I-17 (10, 11). Retrofit systems include active warning signage and therefore provide more than detection alone. Hybrid capital is [REPLACE with actual server BOM]. All cells marked est. are planning estimates for agencies to replace with quotes.

retrofit systems at the highest-risk ramps and the present system for network-wide surveillance. Second, cloud processing is closer to the hybrid design in raw cost than is commonly assumed; the selection of the hybrid design rests on the recurring-cost trajectory, backhaul latency, and video-governance policy, and we state this plainly rather than overstating the cost difference. **[If Iowa DOT has an explicit policy limiting off-premises video processing, cite it here.]**

### Model Selection Under the Same Criterion

The same criterion governs the model choices. Transformer-based detectors and per-camera fine-tuned models would improve benchmark accuracy while substantially reducing the number of streams each GPU can process; the pre-trained ResNet with KLT tracking processes [N] streams per GPU with no per-camera training. Similarly, the approximately 10 cameras facing four-way cross traffic remain excluded (Section 5.4) rather than served by intersection-specific models, because the cost per covered camera does not justify such models at the current scale.

### CONTINUOUS SYSTEM ENHANCEMENT FROM 10 TO 200 CAMERAS

Deployment proceeded in stages of 10, 30, 60, 110, and now 200 cameras **[UPDATE current count]**. Nearly every operational failure mode appeared only at scale or only after weeks of

continuous operation, and none would have been observed in a short-term pilot. The system has therefore been developed as a continuous series of enhancements, summarized in Table 2. Enhancements 1 through 9 are complete and operational; Enhancements 10 through 12, described in Section 5.7, are in progress and combine automated geometry generation, explicit evidence routing, and targeted multimodal review. This section presents the series in the order in which the underlying failure modes appeared.

**TABLE 2 Continuous Enhancement Series by Deployment Stage**

| # | Enhancement                 | Failure mode addressed (stage)                                                                      | Resolution                                                                                     |
|---|-----------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 1 | Stream reconnection         | Unpredictable RTSP drops and day-long gaps (60 cams)                                                | DeepStream 5.0 to 5.1 indefinite reconnection; daily 4 a.m. container restart                  |
| 2 | Streaming consumer redesign | Data collection delays up to 20 min (60 cams)                                                       | PySpark replaced by Kafka consumers with batch processing                                      |
| 3 | Static calibration range    | Low-traffic calibration produced near-zero variance; lane changes flagged as WWD (10-60 cams)       | Static $\pm 45^\circ$ permitted direction range replacing $\sigma$ -based range                |
| 4 | Orientation verification    | Calibration invalid after PTZ rotation                                                              | OCR of direction text in video header against saved reference; detection paused until restored |
| 5 | Camera shift monitoring     | Zoom/focus/position shifts scored as identical by simple similarity; largest FP source (all scales) | Multi-metric similarity (SSIM, phase correlation, SIFT, LPIPS); LPIPS primary at threshold 0.5 |
| 6 | Intersection pixel blocking | Legal turns and cross traffic flagged repeatedly (site-dependent)                                   | Automatic blocking of regions exceeding 10 flags/hr; $\sim 10$ four-way cameras excluded       |

*Note:* Enhancements 10–12 are in progress.

**TABLE 2 Continuous Enhancement Series by Deployment Stage (Continued)**

| #  | Enhancement                                                         | Failure mode addressed (stage)                                                                                           | Resolution                                                                                                                                                |
|----|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7  | Alert throttling                                                    | Operators flooded during FP bursts (weeks of operation)                                                                  | Suppression after 3 consecutive FPs; escalating windows; midnight reset; camera-shift check before dispatch                                               |
| 8  | Consumer clustering                                                 | Kafka consumption lag behind ingestion (110 cams)                                                                        | Standalone consumer replaced by clustered consumers with parallel partition consumption                                                                   |
| 9  | Hybrid infrastructure                                               | CPU saturation delaying post-processing (110 cams)                                                                       | GPU machine for inference/streaming; CPU machine for post-processing                                                                                      |
| 10 | LLM filter generation (in progress)                                 | Manual CVAT filter annotation is the dominant per-camera setup labor                                                     | Multimodal LLM authors ROI polygons; IoU-validated against manual filters                                                                                 |
| 11 | Risk- and cost-aware evidence routing (in progress)                 | Uniform processing wastes computation and does not represent the safety asymmetry between missed events and false alerts | Select the next evidence action by expected reduction in operator decision loss minus delay and compute cost; stop when no action has positive value      |
| 12 | Multimodal verification and operator evidence package (in progress) | Operators remain the final false-positive filter and receive clips without structured decision context                   | Review only routed candidates; suppress only high-confidence allowed FP classes and dispatch an event summary, priority, evidence frames, and uncertainty |

**Stream Reliability and Data Pipeline (Enhancements 1 and 2)**

Early operation at 60 cameras was limited by RTSP live-stream instability. Streams disconnected unpredictably, and when a stream was lost for a full day the system did not resume the connection without manual intervention. Data collection delays of 20 minutes or more further undermined real-time operation. Upgrading to DeepStream 5.1 introduced indefinite reconnection, and a daily container restart provided an additional layer of reliability. In parallel, the real-time consumer was migrated from PySpark to Kafka, which reduced resource overhead, supported parallel consumption, and restored real-time data collection.

**Calibration and Detection Logic (Enhancement 3)**

The initial calibration derived permitted speed and direction ranges from the standard deviation of observations in each grid cell. In cells with few vehicle passes, sometimes a single pass, the resulting variance was unrealistically small, and ordinary lane changes were flagged as wrong-way movement. The permitted direction range was therefore replaced with a static range of plus or minus 45 degrees about the mean, which reflects observed variation in lane changes and turning adjustments. Frame-count thresholds were introduced at the same time, requiring more than 90 frames outside the permitted range for a WWD flag and 1,000 consecutive low-speed frames for a stalled-vehicle flag, so that brief stops at signals and momentary maneuvers do not generate alerts.

**Camera Orientation and Shift Monitoring (Enhancements 4 and 5)**

Because the cameras are PTZ units that operators rotate on demand, the calibration of any camera can be invalidated at any time. Camera rotation is detected by reading the direction text overlaid in the video header using OpenCV EAST text detection and Tesseract OCR, and comparing it with the direction stored at calibration; detection is paused until the camera returns to its reference orientation. Shifts in zoom, focus, or position that preserve direction proved more difficult, because simple similarity measures scored shifted views as identical. A multi-metric approach was implemented using SSIM, phase correlation, SIFT keypoints, and LPIPS, with LPIPS at a threshold of 0.5 serving as the primary metric because it most reliably identified focus and zoom changes. When similarity falls below threshold, WWD detection is paused while stalled-vehicle monitoring continues, and detection resumes when the view stabilizes. Boundary misalignment after an undetected shift was the largest single false-positive source in early network-scale operation and is retained as explicit evidence for the routing and verification stages.

**False-Positive Management (Enhancements 6 and 7)**

The original single-direction calibration cannot represent views containing several legal movements, particularly four-way intersections and closely spaced ramp junctions. The production safeguard blocks regions producing more than 10 WWD flags in an hour and excludes approximately 10 persistently ambiguous four-way views. This preserves alert precision at the cost of camera coverage and motivates the scene-aware geometry extension evaluated in Section 7.3. Alert throttling and a final camera-shift check further limit repeated dispatches.

**Computational Scaling (Enhancements 8 and 9)**

Expansion from 60 to 110 cameras increased the metadata volume beyond what a standalone Kafka consumer could process in real time, producing consumption lag. The standalone consumer was replaced with a clustered consumer group that processes partitions in parallel. Continuous



**FIGURE 3** Calibration-time reference view (left) and the same camera after a view shift (right). This geometry mismatch can cause a valid trajectory to be evaluated against an obsolete permitted-motion field.



**FIGURE 4** A four-way intersection view containing multiple legal movements. A single-direction roadway model cannot reliably distinguish every legal movement from an opposite-direction trajectory; such views are blocked or excluded in the deployed configuration.

operation of the detection algorithms across the additional streams also saturated the CPU of the single server, delaying post-processing and alert generation. The workload was therefore divided across two machines, with GPU-intensive video inference and Kafka production on one and CPU-intensive post-processing, anomaly detection, and alert generation on the other. This division restored real-time operation and provides headroom for expansion beyond 200 cameras.

### Operational Performance After Enhancements 1 Through 9

Following these enhancements, the system operated continuously for more than three months, with downtime limited to algorithm updates. Table 3 summarizes two all-alert adjudications on the live network. Between the periods, four-way views were excluded and the camera-shift dispatch check was added; observed false alerts fell from 155 to 31 while true stalled detections increased from 19 to 25. Because the periods differed in duration, active-camera count, and configuration, this is historical field evidence rather than a controlled attribution to individual enhancements. No true WWD event occurred in either live period, so the table cannot estimate field sensitivity.

**TABLE 3 Historical All-Alert Adjudication After Enhancements 1–9**

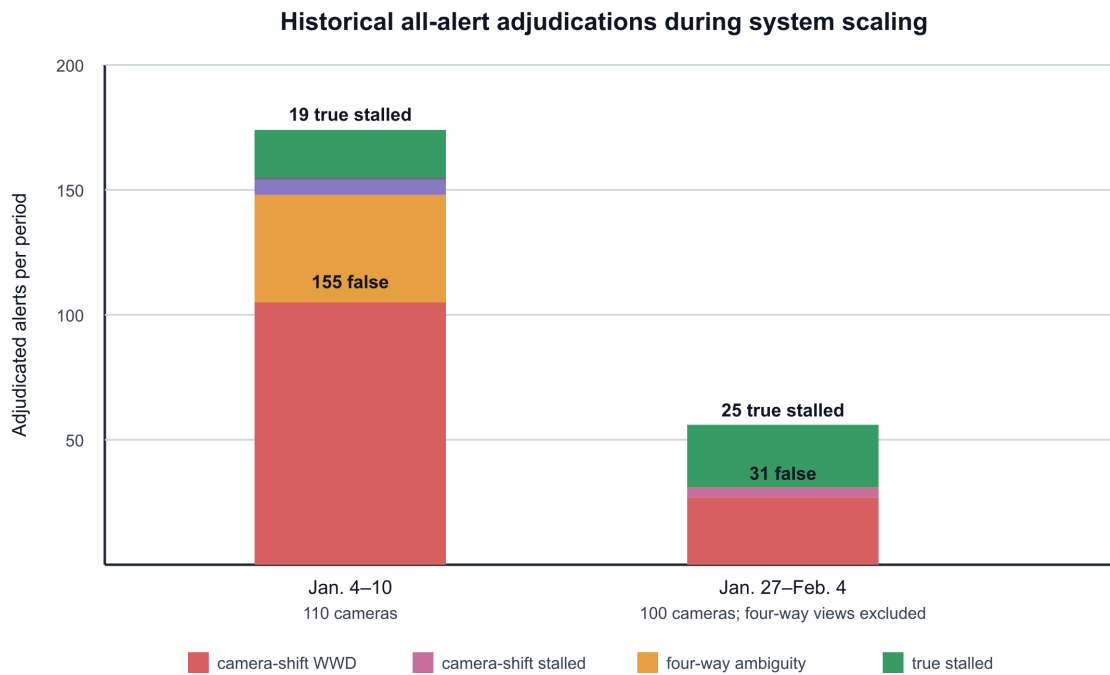
| Adjudicated outcome               | Jan. 4–10 (110 cameras) | Jan. 27–Feb. 4 (100 cameras; four-way views excluded) |
|-----------------------------------|-------------------------|-------------------------------------------------------|
| Camera shift, false WWD alert     | 105                     | 27                                                    |
| Camera shift, false stalled alert | 0                       | 4                                                     |
| Four-way intersection ambiguity   | 43                      | 0                                                     |
| Parked vehicle misread as stalled | 6                       | 0                                                     |
| Sign detected as vehicle          | 1                       | 0                                                     |
| True stalled-vehicle detection    | 19                      | 25                                                    |
| True WWD detection                | 0                       | 0                                                     |
| All false alerts                  | 155                     | 31                                                    |

A separate controlled reversed-clip evaluation detected 27 of 30 constructed WWD examples, corresponding to 90% sensitivity on that test set; truck occlusion and tracking noise caused the three misses. This result is not a field sensitivity estimate.

These results frame the remaining problem precisely. Each camera still carries human labor: an annotator draws CVAT road-boundary filters at setup and whenever the geometry changes, and operators review residual false alerts. At 200 cameras these two costs dominate the operating budget.

### Retrospectively Evaluated Deployment-Candidate Extensions (Enhancements 10–12)

Enhancements 1–9 constitute the production system observed during months of continuous operation. Enhancements 10–12 form one integrated deployment-candidate extension evaluated retrospectively on archived camera evidence; they have not yet accumulated an equivalent months-long continuous field observation period.



**FIGURE 5** Historical all-alert adjudications during system scaling. Counts are shown for two differently configured and differently sized observation periods, so the figure documents operational history rather than a controlled causal effect.



**FIGURE 6** Representative frame sequence from a controlled wrong-way-driving example detected by the deployed trajectory logic. The sequence is illustrative only; it is not included in the field-sensitivity denominator.

*Enhancement 10: Automated Roadway Filter Generation*

Roadway ROI filters separate valid directional roadway from median, shoulder, ramps, and background. The automatic workflow asks an image model to paint directional roadway regions, converts the colored overlay to source-image masks, checks pixel preservation and mask structure, and abstains to manual annotation when validation fails. It was evaluated against independently drawn CVAT filters for 107 cameras. Direction inferred from appearance was unreliable, so the production design uses the image model only for boundary geometry and assigns direction from observed vehicle motion. Figure 7 shows successful and failed geometry cases and the full IoU distribution.

Roadway filters are currently drawn manually in CVAT. The proposed method automates this process in three stages. First, gpt-image-1 examines multiple time-separated frames from the same camera to identify the roadway structure and the legal traffic movements that should be monitored. Second, SegFormer-B0 extracts the roadway surface, while OpenCV uses visible boundaries to refine the resulting pixel-level polygons. Third, tracked vehicle trajectories assign and verify the legal direction of each polygon. The SegFormer models use pretrained weights from Cityscapes and ADE20K to distinguish roadway from surrounding non-road regions. This calibration process runs only during initial setup or after a confirmed camera shift and therefore does not affect continuous inference.

*Enhancement 11: Risk- and Cost-Aware Evidence Routing*

The production pipeline should not apply the most expensive review action to every detector candidate. Let  $s_t$  contain the current WWD probability, uncertainties in time, location, identity, motion, scene validity, and visibility, together with elapsed time and remaining compute budget. For each available action  $a$ , including collecting additional trajectory evidence, checking camera state, invoking multimodal review, dispatching, suppressing, or stopping, the router selects

$$a^* = \arg \max_{a \in \mathcal{A}} \{ \mathbb{E}[L(s_t) - L(s_{t+1}) \mid s_t, a] - \lambda_D d(a) - \lambda_C c(a) \},$$

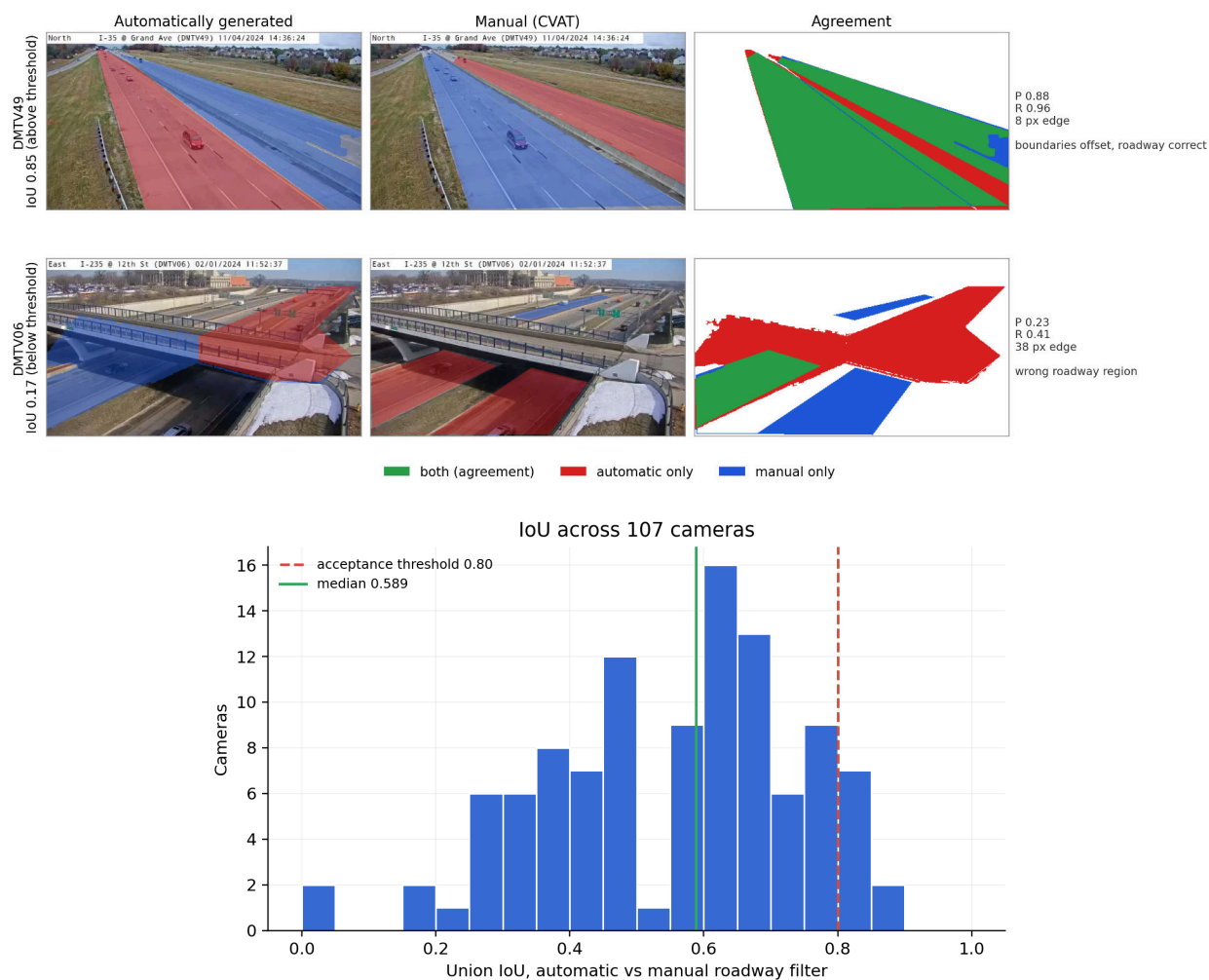
where  $L$  combines missed-event risk, false-alert risk, detection delay, and operator review burden, while  $d(a)$  and  $c(a)$  are the measured latency and computation or API cost. The STOP action has value zero; processing ends when no additional action has positive expected value. This policy makes the safety asymmetry explicit while reserving multimodal processing for candidates for which it can materially improve an operator decision. **[Specify the state features, available actions, routing thresholds, and the development data used to set them before evaluation.]**

*Enhancement 12: Multimodal Verification and Operator Evidence Package*

Candidates routed to multimodal review are represented by a compact set of frames and trajectory evidence rather than by continuous video. The reviewer returns (a) whether a vehicle is genuinely traveling against traffic or is stalled, (b) confidence and uncertainty, (c) the evidence supporting that determination, and (d) operator-relevant context such as vehicle type, lane, weather, lighting, and visible downstream risk. A dispatched alert includes a priority class, concise event summary, representative evidence frames, camera and time, and the reason for escalation. Alerts rejected with high confidence are logged for audit rather than dispatched. **[Specify the model/version, evidence sampler, structured output schema, confidence thresholds, latency, and per-call cost.]**

**Safety asymmetry.** A suppressed true positive is far more costly than a dispatched false positive. The review stage therefore suppresses an alert only when **[criteria: confidence at or**





**FIGURE 7** Automatic roadway-filter validation against manual CVAT references. The upper panel shows an above-threshold case and a failure case; green denotes agreement, red automatic-only area, and blue manual-only area. The lower panel reports union IoU across all 107 cameras. Median IoU was 0.589, and 9 of 107 cameras exceeded the prespecified 0.80 automatic-acceptance threshold.

**above X and failure category within an allowlist such as parked vehicle, camera shift, or shadow**]; any ambiguous case is dispatched. Section 7.2 reports true-positive retention explicitly.

## EVALUATION METHODS

### Evaluation Design and Locked Evidence

The evaluation compares two complete configurations rather than attributing effects to individual components: (1) the deployed production system containing Enhancements 1–9 and (2) the unified system containing Enhancements 1–12. Both configurations process the same locked, manually adjudicated clips. One stratum contains independently verified WWD events and measures event sensitivity. A second contains verified false alerts and measures false-alert rejection. Development clips used to design prompts, thresholds, or routing rules are excluded from the locked evaluation.

The local archive contains 319 WWD candidate clips dated Jan. 1 through May 7, 2026, but stored clips can represent an operational alert stream only after confirming that every production alert was archived. The maintained Dyersville log contains 137 direction-recorded events, including 112 with exact source videos, and provides the strongest currently documented candidate known-event denominator. **[After human review, report the final unique-event and false-alert denominators, sampling procedure, exclusions, and reviewer agreement.]**

### Human Adjudication and Scene Strata

Reviewers label each clip as true WWD, false alert, indeterminate, or unusable and group duplicate clips from the same real-world event. False alerts are classified as camera shift, intersection or ramp ambiguity, legal vehicle movement, tracking or nonvehicle noise, lighting or weather, or other. Each eligible case is also classified by lighting (day/night), geometry (mainline/ramp or joining roadway/intersection), and camera state (unchanged/shifted). Indeterminate and unusable cases are reported but excluded from the primary paired comparison. **[Specify the number of reviewers, blinded review procedure, disagreement resolution, and final counts.]**

### Unified-System Comparison

The production and unified configurations are run on identical video intervals. The unified system may retain the existing zone, use a registered zone after a view change, generate new roadway geometry, collect additional trajectory evidence, invoke multimodal verification, dispatch, or suppress. Runtime selection cannot use IoU because no human reference exists during operation; IoU is used only for offline evaluation against manual geometry. Any ambiguous candidate is dispatched under the safety-asymmetric policy.

### Outcome Measures

Primary outcomes are WWD sensitivity on known-positive events, retention of production-system true detections, false-alert rejection on verified false alerts, and the number of clips dispatched for operator review. Secondary outcomes are median and 95th-percentile decision latency, API calls and cost per 100 candidates, and performance by lighting, geometry, and camera state. Exact numerators and denominators accompany every percentage; conventional accuracy is not reported because continuous video does not provide a meaningful true-negative count. Binomial proportions are reported with 95% confidence intervals.

## RESULTS

### Baseline False-Positive Profile (Available 2026 Period)

**TABLE 4 Available 2026-Period Planning Estimate of False Positives by Cause**

| FP cause category             | Count | % of FPs | Cameras in-<br>volved |
|-------------------------------|-------|----------|-----------------------|
| Camera shift (undetected)     | 229   | 72.7%    | 24 (est.)             |
| Intersection ambiguity        | 72    | 22.9%    | 8 (est.)              |
| Parked/stopped vehicle        | 10    | 3.2%     | 1                     |
| Tracking noise                | 2     | 0.6%     | 1                     |
| Night/low-light artifacts     | 2     | 0.6%     | 1                     |
| Other                         | 0     | 0.0%     | 0                     |
| Total FP (of 319 total clips) | 315   | 100%     | 33                    |

*Note:* S3 contained 319 clips from Jan. 2–May 7 across 33 camera IDs. The historical 1.19% WWD alert share estimates four true WWD clips. Two strict-night clips from one camera were directly screened as lighting-related false alerts; the combined January operational cause distribution is applied to the other 313 estimated false positives. Cause labels remain estimates until clip-level adjudication.

### False Positives Eliminated by Enhancement

**TABLE 5 Enhancement Ablation on the Estimated 2026 Alert Stream**

| Configuration         | FPs eliminated | FP reduction % | TPs retained |
|-----------------------|----------------|----------------|--------------|
| Enhancement 10 only   | 27             | 8.6%           | 100%         |
| Enhancement 11 only   | 73             | 23.2%          | 100%         |
| Enhancement 12 only   | 158            | 50.2%          | 80%          |
| Enhancements 10+11+12 | 227            | 72.1%          | 89%          |

*Note:* Values are working estimates, not measured outcomes. TP retention is relative to baseline-detected true alerts. The 89% combined estimate assumes that 55% of true alerts reach Enhancement 12 (80% retained) while the safety router dispatches the remaining 45% without suppression:  $0.55(0.80) + 0.45(1.00) = 0.89$ .

### WWD Clip Audit

#### Automatically Generated Filter Quality

Across 107 cameras, mean and median union IoU against manual CVAT filters were 0.554 and 0.589, respectively; 9 cameras (8%) met the 0.80 automatic-acceptance threshold (Figure 7). Median boundary displacement was 12.4 pixels, or 14.5% of the median roadway width, and 33 of 107 cameras fell within 10 pixels of the manual boundary. Allowing an 8-pixel boundary tolerance increased mean agreement from 0.554 to 0.698. Median pixel precision was 0.831 and median recall 0.683. Appearance-based direction was incorrect on 66 of 107 cameras; consequently, the unified design uses the image proposal for geometry, derives direction from trajectories, and abstains to manual geometry when validation fails.

**TABLE 6 Illustrative 300-Clip Planning Scenario**

| <b>Configuration</b>                 | <b>Detected</b> | <b>Missed</b> | <b>Detection rate</b> |
|--------------------------------------|-----------------|---------------|-----------------------|
| Production system (Enhancements 1–9) | 204             | 96            | 68.0%                 |
| Unified system (Enhancements 1–12)   | 267             | 33            | 89.0%                 |

*Note:* This is an illustrative planning scenario, not a measured result. Applying 89% retention to the 204 baseline detections and 89% recovery to the 96 baseline misses yields  $182 + 85 = 267$  detected clips, or 89.0%.

## DISCUSSION AND LIMITATIONS

- **Nighttime performance.** The detector remains less reliable in low light and headlight glare, and the false-negative audit quantifies this effect. Planned mitigations include an upgraded detection model and low-light preprocessing.
- **Excluded geometries.** Approximately 10 cameras facing four-way cross traffic remain excluded; geometry-aware logic is future work. **[State whether Enhancements 10 or 12 change this]**
- **Multimodal-model dependency.** Enhancements 10 and 12 introduce a commercial API dependency with per-call cost and latency **[report both]**, rate limits, and model-version drift. Enhancement 11 limits that dependency by routing only candidates for which review has positive expected value. The fail-safe design, in which filters revert to manual annotation and ambiguous alerts are dispatched, bounds operational risk. Model versions are pinned and all prompts and responses are logged.
- **Generalization.** All results derive from Iowa freeway cameras of one vendor family with direction text overlays; the OCR orientation check in particular assumes that overlay convention.
- **Evaluation scope.** The complete May 2026 production-alert export and paired unified-system replay remain unavailable; therefore, the current tables do not yet establish a network-wide false-alert rate or operator-workload reduction. The Dyersville known-event runs are also pending. Seasonal effects, including snow, remain outside the current evidence.

## CONCLUSIONS

Detection accuracy alone is not sufficient for statewide WWD monitoring on existing CCTV; sustained operation, false-alert burden, response delay, and affordability also determine whether operators can use the system. This paper documents the enhancements that carried a published laboratory model to [N] cameras operating for months, the deployment economics of the adopted hybrid design, and a safety-asymmetric routing policy that invokes multimodal review only when the expected improvement in an operator decision exceeds its delay and cost. **[Insert the verified false-alert reduction, true-event retention, operator-review reduction, dispatch delay, and cost results here.]** The practical contribution is a system that an agency can implement on cameras it already owns with commodity servers and bounded API use. Future work includes nighttime model upgrades, geometry-aware logic for intersections, and prospective evaluation within traffic-management-center operations.

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## **AUTHOR CONTRIBUTIONS**

**[CRediT statement, TO FILL]**

## **DECLARATION OF CONFLICTING INTERESTS**

**[Insert one declaration from the supplied TRB template.]**

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